

Source-specific pollution exposure and associations with pulmonary response in the Atlanta Commuters Exposure Studies

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Contents

1	Part A. Details about the PMF model	1
1.1	Constraints in the PMF model	1
1.2	Rescaling PMF results	3
2	Tables	3
2.1	Table S1. Units, mean (minimum, maximum) amount, and missingness among pollutants used to estimate source-specific traffic pollution.	3
2.2	Table S2. Correlation across commutes between concentrations of source-specific traffic pollution.	4

1 Part A. Details about the PMF model

1.1 Constraints in the PMF model

To develop the auxiliary equations Q^{aux} in constrained PMF, we first determined pollutants associated with each of the four sources (crustal, non-tailpipe dominant, primary tailpipe dominant, secondary), and pollutants not associated with each source. The pollutants associated with each source l should be reflected in its source profile $\mathbf{f}_l$. Therefore, each auxiliary equation Q_k^{aux} constrains f_{lp} for some source l and constituent p , where $Q^{aux} = \sum_{k=1}^K Q_k^{aux}$. These constraints Q_k^{aux} are of the form

$$\frac{(f_{lp} - a_{lp})^2}{s^2}$$

where s controls the strength of the penalty and a_{lp} is the target value that f_{lp} should be close to. However, in practice a_{lp} may be unknown, and so the ME-2 approach also can “pull” each f_{lp} to be larger or smaller than its unconstrained value by using inequality constraints with a limit on the total Q_k^{aux} for each constraint k [2, 3]. Because we did not have target values a_{lp} in our application, we applied these inequality constraints.

Briefly, we increased aluminum, calcium, and noise in the crustal source; zinc and iron in the non-tailpipe dominant traffic source; and OC and WSOC in our secondary source. We decreased amounts of WSOC in crustal; aluminum and calcium in the secondary source; and OC and BC in the primary tailpipe dominant source. These auxiliary equations helped to better separate sources that emit similar components. The auxiliary equations are summarized in the following table:

Source	Increased	Decreased
Crustal	Al, Ca, noise	WSOC
Non-tailpipe dominant	Fe, Zn	
Primary tailpipe dominant		OC, BC
Secondary	OC, WSOC	Al, Ca

1.2 Rescaling PMF results

The raw source-specific pollution matrix $\mathbf{G}$ from PMF is unitless. In order to rescale results to be more interpretable, we regressed observed total $\text{PM}_{2.5}$ on the source-specific estimates from PMF, $\hat{g}_{il}$,

$$PM_i = \sum_{l=1}^L \beta_l \times \hat{g}_{il} + \epsilon_i$$

where PM_i represents $\text{PM}_{2.5}$ for commute i and $\hat{g}_{il}$ represents the estimated source-specific pollution for commute i and source category l from PMF. Then, we estimated source-specific pollution concentrations in $\mu\text{g}/\text{m}^3$ using $g_{il} = \beta_l \times \hat{g}_{il}$. For each commute, the sum of contributions in $\mu\text{g}/\text{m}^3$ from each source should be approximately equal to the total $\text{PM}_{2.5}$ in $\mu\text{g}/\text{m}^3$. This regression approach for scaling has been previously proposed [1].

2 Tables

2.1 Table S1. Units, mean (minimum, maximum) amount, and missingness among pollutants used to estimate source-specific traffic pollution.

pollutant	units	mean	(minimum, maximum)	N missing	% missing
BC	$\mu\text{g}/\text{m}^3$	5.64	(0.97, 15.89)	6	3.7
OC	$\mu\text{g}/\text{m}^3$	12.83	(0.11, 38.45)	7	4.3
WSOC	$\mu\text{g}/\text{m}^3$	4.87	(0.2, 39.2)	0	0.0
PNC	n/cm^3	21951.28	(4346.96, 68950.94)	15	9.3
pbPAH	ng/m^3	115.07	(30.03, 206.57)	23	14.2
B	ng/m^3	3.94	(0.02, 23.29)	0	0.0
S	ng/m^3	298.93	(6.85, 2784.77)	0	0.0
As	ng/m^3	0.48	(0.01, 3.32)	0	0.0
Pb	ng/m^3	1.25	(0.02, 16.66)	0	0.0
Al	ng/m^3	36.27	(0.41, 264.88)	0	0.0
Ca	ng/m^3	47.52	(0.77, 374.73)	0	0.0
K	ng/m^3	27.62	(1.36, 132.59)	0	0.0
Ba	ng/m^3	21.77	(0.06, 124.74)	0	0.0
Ni	ng/m^3	1.27	(0.06, 23.09)	1	0.6
V	ng/m^3	0.49	(0.01, 7.12)	0	0.0
Ce	ng/m^3	0.13	(0, 2.34)	0	0.0
Fe	ng/m^3	236.32	(0.52, 1358.87)	0	0.0
Zn	ng/m^3	12.77	(0.14, 170.36)	0	0.0
Mn	ng/m^3	2.21	(0.01, 12.02)	0	0.0
Cr	ng/m^3	1.41	(0.01, 7.38)	0	0.0
Mg	ng/m^3	9.10	(1.13, 39.44)	0	0.0
U	ng/m^3	0.00	(0, 0.02)	0	0.0
P	ng/m^3	5.31	(0.05, 26.11)	0	0.0
Ti	ng/m^3	9.66	(0.03, 55.14)	0	0.0
Cd	ng/m^3	0.03	(0, 0.25)	0	0.0
Co	ng/m^3	0.04	(0, 0.17)	0	0.0
Sb	ng/m^3	2.83	(0.02, 18.44)	0	0.0
Noise	dB	70.04	(62, 81.94)	20	12.3

2.2 Table S2. Correlation across commutes between concentrations of source-specific traffic pollution.

	Crustal	Non-tailpipe dominant	Primary tailpipe dominant	Secondary
Crustal	1.00	0.74	-0.46	0.54
Non-tailpipe dominant	0.74	1.00	-0.23	0.58
Primary tailpipe dominant	-0.46	-0.23	1.00	-0.23
Secondary	0.54	0.58	-0.23	1.00

References

- [1] Eugene Kim, Philip K. Hopke, Joseph P. Pinto, and William E. Wilson. Spatial Variability of Fine Particle Mass, Components, and Source Contributions during the Regional Air Pollution Study in St. Louis. *Environmental Science & Technology*, 39(11):4172–4179, June 2005.
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- [3] Pentti Paatero and Philip K Hopke. Rotational tools for factor analytic models. *Journal of Chemometrics*, 23(2):91–100, 2009.